STANDARD OPERATING PROCEDURE (SOP)

Network Troubleshooting Procedures for SOC.lab

Title	Network Troubleshooting Procedures for SOC.lab	**SOP ID**	NET-OPS-TS-001
Version	1.0	**Status**	Approved Draft
Environment	SOC.lab	**Author**	Joshua Sartain
Department	Network Engineering / Security Operations	**Document Type**	Operational Troubleshooting SOP

Purpose Summary: This SOP standardizes troubleshooting for network and telemetry failures directly related to the SOC.lab stack, including OPNsense, Hyper-V virtual switching, managed switch infrastructure, VLANs, Windows Server / Active Directory services, RDS access, WAN connectivity, and Wazuh-related telemetry paths. The document is organized by symptom, affected zone, and recovery path so operators can move from rapid triage to validated service restoration without making unsafe changes during an outage.

1. Purpose

This SOP defines the standard troubleshooting workflow for network-related incidents within SOC.lab. It is intended to help analysts, infrastructure administrators, and network/security engineers quickly identify, isolate, remediate, validate, and document failures affecting user access, inter-segment communication, core services, and security telemetry.

The troubleshooting model in this SOP is built around the actual SOC.lab architecture, not a generic enterprise network. It assumes segmented internal zones, mixed physical and virtual switching, OPNsense as a major routing and policy point, Windows Server / Active Directory dependence for DNS and DHCP, and Wazuh-related telemetry paths that must be preserved during diagnosis and recovery.

2. Scope

This SOP applies to all network troubleshooting activities involving the following SOC.lab systems and service paths:

- OPNsense firewall, routing, and interface assignments
- Hyper-V host networking, including vSwitch, VM NIC, and physical NIC bindings
- Managed switches, trunk ports, access ports, and VLAN forwarding paths
- SOC.lab network zones and subnets, including user, management, server, SIEM, WAN, and lab segments
- Windows Server / Active Directory dependencies, including DNS and DHCP
- RDS access paths and thin-client connectivity
- Wazuh and related telemetry transport paths, including syslog, agent communication, and firewall log forwarding
- Internet/WAN availability and upstream connectivity paths

3. SOC.lab Troubleshooting Boundaries

The following names and architectural boundaries shall be used throughout troubleshooting:

Zone / Service	Primary Function	Typical Failure Impact
SOC-MGMT	Management and administrative traffic	Loss of admin control, reduced troubleshooting capability
SOC-LAN	Core internal workstation and analyst operations network	User and analyst service interruption
SOC-SERVER	Server-hosted infrastructure and service workloads	AD, RDS, application, and file service disruption
SOC-SIEM	Monitoring and logging systems	Reduced visibility, alerting, and evidence continuity
SOC-USER	Thin-client or user-side access path	Client access failures and session loss
SOC-WAN	External or upstream connectivity path	Internet outage, update failure, remote reachability loss
SOC-LAB	Testing and isolated research workloads	Contained service failure or false spillover into production-like segments

4. Roles and Responsibilities

- SOC Analyst: identifies symptoms, preserves evidence, checks dashboards and service health, and escalates infrastructure-impacting issues.
- Infrastructure Administrator: validates Hyper-V, Windows Server, RDS, and virtual switch conditions; assists with host-side remediation.
- Network Engineer: validates switch state, VLANs, trunk/access port behavior, routing assumptions, and physical network conditions.
- Security Engineer: validates OPNsense policy effects, firewall logging paths, and network telemetry continuity into Wazuh.
- System Owner / Lead Engineer: authorizes higher-risk remediation, approves changes that may affect architecture or trust boundaries, and reviews outage documentation.

5. Severity Classification

Severity	Definition	Examples in SOC.lab	Escalation Expectation
Critical	Core security or identity services unavailable across multiple zones	OPNsense down, AD/DNS unavailable, Hyper-V uplink misbound, major WAN outage	Immediate escalation to infrastructure and network/security engineering
High	Major user or server impact in one or more critical segments	RDS unreachable, DHCP failure across user zone, trunk/VLAN failure affecting multiple systems	Escalate within active response window
Medium	Partial service degradation, telemetry loss, or zone-limited failure	Wazuh log forwarding failure, partial internet access, one VLAN isolated	Troubleshoot and document, escalate if unresolved
Low	Limited or isolated issue with low blast radius	Single endpoint wrong subnet, one stale static route, one switchport	Resolve locally and document

		mispatch	

6. Administrative Access Assumptions

The following access paths may still be available during a network outage and should be prioritized in the order below:

- Local console or direct access to the affected system
- Hyper-V console access for affected virtual machines and supporting infrastructure
- OPNsense local or VM console access
- Switch local CLI or management access through an unaffected management path
- Radmin or equivalent remote administrative path where still functional
- RDS administrative sessions if identity, routing, and host access remain operational

7. Prohibited Troubleshooting Actions

The following actions shall not be performed without explicit approval from the system owner or designated lead engineer:

- Resetting OPNsense to defaults or applying unvalidated configuration imports
- Rebooting the Hyper-V host as a first-response action
- Changing production VLAN IDs, native VLAN assignments, or trunk allowed lists blindly
- Deleting or broadly disabling firewall rules without documenting the reason and expected effect
- Modifying AD-integrated DNS or DHCP scope settings without verifying the dependency chain
- Introducing duplicate IP ranges, temporary static addresses, or emergency adapters without recording them
- Disabling logging, SIEM forwarding, or security monitoring except as part of an approved containment or maintenance action

8. Common Known Root Causes in SOC.lab

- Wrong Hyper-V vSwitch binding, especially internal vs external switch misuse
- Duplicate subnets on isolated segments that appear logically similar but do not share the same Layer 2 path
- Access port and trunk port mismatch between switch and upstream/downstream device
- Incorrect VLAN membership on edge ports or missing VLAN allowance on trunk paths
- Disabled or failed physical NIC on Hyper-V host or server
- OPNsense interface misassignment following host or virtual NIC changes
- WAN DHCP failure or upstream path failure on the OPNsense external interface
- AD/DNS unavailability causing apparent network failure even when L3 reachability exists
- DHCP overlap between Windows Server and another unintended source
- Firewall rule block or NAT path issue causing partial connectivity symptoms
- Telemetry misbinding, log forwarding breakage, or agent-to-manager path failure affecting Wazuh visibility

9. Standard Troubleshooting Workflow

9.1 Identify the Symptom

Document the exact user-visible symptom first. Examples include no internet access, no gateway reachability, successful internet access but no internal service access, failed DNS resolution, failed RDS login path, no SIEM telemetry, or complete segment isolation.

9.2 Define the Blast Radius

Determine whether the issue affects one endpoint, one VLAN, one service, one zone, or multiple dependent systems. Note whether only user traffic is affected, only management traffic is affected, or whether monitoring/telemetry paths are also degraded.

9.3 Verify Dependency Chain

Check identity, DNS, gateway, route path, and firewall dependencies before making changes. SOC.lab service failures often cascade because Active Directory, DNS, OPNsense, Hyper-V networking, and Wazuh telemetry all depend on one another.

9.4 Validate the Most Probable Failure Point

Use the symptom-to-root-cause matrix in this SOP to test the most likely issue before moving outward. Start with local link and addressing, then VLAN/port role, then gateway/routing, then DNS/AD service path, then application-specific reachability.

9.5 Restore Minimum Service Safely

If remediation is needed, use the smallest change required to restore service. Avoid architecture changes during troubleshooting. Record what changed, who approved it, and what validation was performed afterward.

9.6 Validate Recovery

Confirm that the original symptom is resolved, dependent services are restored, and monitoring/logging remain intact. Recovery is not complete until user path, service path, and telemetry path are all validated where applicable.

10. Symptom-Based Troubleshooting Procedures

10.1 No Connectivity from an Endpoint

Examples: endpoint cannot ping gateway, cannot reach local server, and cannot reach internet.

- Confirm link status on the endpoint NIC and switch port.
- Validate IP address, subnet mask, default gateway, and DNS assignment.
- Verify the endpoint is on the expected switch port and correct access VLAN.
- Check whether the switch port is administratively up and not err-disabled.
- Confirm the default gateway interface is up on OPNsense or the routed device for that segment.
- Validate whether DHCP lease assignment came from Windows Server / AD or whether the endpoint fell back to stale static values.
- If multiple systems in the same zone fail, test the upstream switch, trunk, and gateway path rather than the endpoint alone.

10.2 Partial Connectivity

Examples: endpoint can reach some internal systems but not internet, or internet works but internal services do not.

- If internet works but internal services do not, test DNS and AD-integrated service resolution first.
- If the gateway responds but upstream routes fail, inspect OPNsense policy, NAT, and route path.
- If only internal servers are unreachable, confirm VLAN-to-subnet mapping and inter-VLAN routing assumptions.

- If only one class of traffic fails, inspect firewall rules or ACL behavior rather than basic switching.
- Confirm whether the issue is name resolution only by testing direct IP reachability versus FQDN reachability.

10.3 Wrong Subnet or Gateway Assignment

Examples: endpoint receives the wrong range, duplicate gateway behavior, or no correct lease.

- Review DHCP scope, reservations, relay assumptions, and exclusions on Windows Server / AD.
- Confirm no second DHCP server is active on that segment.
- Verify the switch port access VLAN and any Hyper-V virtual NIC network attachment for virtual endpoints.
- Release and renew the lease only after validating the expected scope.
- Document any rogue or overlapping DHCP behavior as a security issue as well as an availability issue.

10.4 Can Ping Gateway but Not Internet

Examples: local gateway responds but external addresses fail.

- Check OPNsense WAN interface status and upstream IP assignment.
- Confirm WAN DHCP or static configuration is still valid.
- Inspect default route presence on OPNsense.
- Test whether DNS is the only failing component by pinging a known external IP and then resolving a name.
- Check upstream managed switch connectivity and physical WAN handoff.
- Inspect firewall/NAT rules if the WAN is up but traffic is not translating.

10.5 Can Reach Internet but Not Internal Services

Examples: browsing works but AD, file services, RDS, or Wazuh services are unreachable.

- Validate DNS server availability on Windows Server / AD.
- Test internal FQDN resolution and direct IP resolution separately.
- Confirm that the relevant server VLAN/subnet is reachable and not isolated by wrong switchport role or missing trunk allowance.
- Inspect OPNsense policy between the source zone and the target internal service zone.
- If RDS is affected, confirm the server is reachable locally and that AD authentication is available.

10.6 DNS Resolution Failure

Examples: can ping by IP but not by name; AD logon and service discovery fail.

- Confirm the client is pointed to the correct DNS server(s), typically Windows Server / AD and approved forwarder path.
- Validate DNS service status on the server.
- Check whether the OPNsense DNS forwarder path or upstream forwarders have failed.
- Test lookup of local zone records, then external names, to isolate whether failure is internal-only or upstream.
- Review AD/DNS event logs if authentication-related name resolution issues appear simultaneously.

10.7 Active Directory / Authentication Related Network Symptom

Examples: network appears up, but domain logon, GPO, name resolution, or RDS auth fails.

- Validate domain controller reachability by IP and hostname.
- Check DNS first; most AD-related network failures in SOC.lab present as DNS failures before they appear as authentication failures.

- Confirm time synchronization is not causing authentication rejection.
- Verify required service ports are not blocked between source and domain controller segments.
- If only RDS users are affected, test direct DC path from the RDS host.

10.8 RDS Unreachable or Session Path Failure

Examples: thin clients cannot reach RDS-SRV01 or session establishment fails.

- Test IP reachability to the RDS host.
- Validate name resolution for the RDS host.
- Confirm the RDS server still has correct network assignment in Hyper-V and that the corresponding vSwitch is bound correctly.
- Check whether AD authentication is available.
- Confirm host firewall and network path are both functioning.
- If only thin clients are affected, validate their VLAN and gateway assignment separately from the RDS host itself.

10.9 VLAN Isolation Broken or Cross-Zone Leakage

Examples: zones can talk when they should not, or a VLAN cannot reach only expected routed destinations.

- Inspect access VLAN assignment on affected edge ports.
- Review trunk allowed VLAN list and native VLAN assumptions.
- Validate OPNsense interface-to-zone mapping.
- Confirm there is no duplicate subnet deployed across isolated segments.
- Check for accidental bridge behavior or a vSwitch attached to the wrong physical NIC.
- Treat unexpected cross-zone communication as a security event and document both availability and policy impact.

10.10 Hyper-V Virtual Network Issue

Examples: VMs lose connectivity after host changes, or one virtual network behaves differently from another.

- Map the VM NIC to the intended vSwitch.
- Validate whether the vSwitch is internal or external and whether that is intentional.
- Confirm the physical NIC bound to the external vSwitch is the correct one and is linked upstream.
- Check whether management OS sharing was enabled or disabled intentionally.
- If the VM subnet overlaps another isolated segment, verify that the issue is not duplicate addressing across non-shared Layer 2 domains.

10.11 OPNsense WAN or Interface Failure

Examples: no external connectivity, one zone loses routing, or firewall interface names changed after host-side updates.

- Check interface assignments against the expected SOC.lab zone design.
- Validate link state and upstream connectivity.
- Confirm gateway presence and route table health.
- Review firewall logs for unexpected deny behavior.
- Check whether a Hyper-V NIC order or host-side adapter change altered interface mapping inside the OPNsense VM.

10.12 Wazuh / Telemetry Connectivity Failure

Examples: firewall logs stop, agents stop reporting, syslog feed breaks, or SIEM visibility becomes incomplete.

- Confirm source systems still have network connectivity to the Wazuh manager or log collector path.
- Validate local agent status on the endpoint or server.
- Test reachability from OPNsense, switches, and monitored systems toward the telemetry destination.
- Check whether DNS, routing, or firewall rules changed on the telemetry path.
- If network service is restored but telemetry remains broken, restart only the specific forwarding service rather than broad platform services.
- Document telemetry outage start time and recovery time for evidence continuity.

11. Core Troubleshooting Commands and Tools

Tool / Command	Use in SOC.lab	Notes
ping / traceroute / tracert	Basic path and reachability validation	Use both internal targets and gateway/WAN targets
ipconfig /all	Windows endpoint addressing and DNS validation	Confirm DHCP source, lease details, and DNS order
nslookup	Internal and external DNS troubleshooting	Test both direct DNS server and default server behavior
PowerShell Test-NetConnection	Protocol-specific connectivity checks	Useful for RDS, Wazuh, and service port checks
arp / route print	Local neighbor and route inspection	Helpful for wrong gateway or duplicate path symptoms
OPNsense console / web UI	Interface state, gateway, routing, and firewall log validation	Use console when management path is impaired
Hyper-V Manager	vSwitch, VM NIC, host NIC, and adapter binding validation	Check internal vs external switch type carefully
Switch CLI	Port state, VLAN membership, trunk status, err-disable, interface counters	Critical for trunk/access mismatch investigation
Wazuh agent status / manager checks	Telemetry path validation	Use after network path is confirmed stable

12. Validation Requirements After Remediation

- Original symptom resolved and reproduced validation no longer fails
- Correct subnet, gateway, and DNS settings present on affected endpoints or servers
- Expected internal services reachable
- Expected internet or WAN path restored where applicable
- Security telemetry restored to Wazuh or other monitoring platforms when affected
- No unintended cross-zone communication introduced during remediation
- All changes logged with time, approver, operator, and validation evidence

13. Documentation and Evidence Retention

Every network incident handled under this SOP shall include, at minimum:

- Date and time the issue was detected
- Affected zone, device, and user/service impact

- Exact symptom description
- Tests performed and results
- Root cause determination
- Change(s) made during remediation
- Validation evidence after recovery
- Whether telemetry continuity was affected
- Operator name and reviewer / auditing space

14. Auditing and Sign-Off

Name	Role	Signature / Date
Joshua Sartain	Author & Lead engineer	JS 12/15/2025
Remmington Chamber	Technical Review / Auditing	RC 12/15/2025